

Lesson 2 : LangGraph Components

```
In [1]: from dotenv import load_dotenv
_ = load_dotenv()
```

```
In [2]: from langgraph.graph import StateGraph, END
from typing import TypedDict, Annotated
import operator
from langchain_core.messages import AnyMessage, SystemMessage, HumanMessage,
from langchain_openai import ChatOpenAI
from langchain_community.tools.tavily_search import TavilySearchResults
```

```
In [3]: tool = TavilySearchResults(max_results=4) #increased number of results
print(type(tool))
print(tool.name)
```

```
<class 'langchain_community.tools.tavily_search.tool.TavilySearchResults'>
tavily_search_results_json
```

If you are not familiar with python typing annotation, you can refer to the [python documents](#).

```
In [4]: class AgentState(TypedDict):
        messages: Annotated[list[AnyMessage], operator.add]
```

Note: in `take_action` below, some logic was added to cover the case that the LLM returned a non-existent tool name. Even with function calling, LLMs can still occasionally hallucinate. Note that all that is done is instructing the LLM to try again! An advantage of an agentic organization.

```
In [5]: class Agent:

    def __init__(self, model, tools, system=""):
        self.system = system
        graph = StateGraph(AgentState)
        graph.add_node("llm", self.call_openai)
        graph.add_node("action", self.take_action)
        graph.add_conditional_edges(
            "llm",
            self.exists_action,
            {True: "action", False: END}
        )
        graph.add_edge("action", "llm")
        graph.set_entry_point("llm")
        self.graph = graph.compile()
        self.tools = {t.name: t for t in tools}
        self.model = model.bind_tools(tools)
```

```

def exists_action(self, state: AgentState):
    result = state['messages'][-1]
    return len(result.tool_calls) > 0

def call_openai(self, state: AgentState):
    messages = state['messages']
    if self.system:
        messages = [SystemMessage(content=self.system)] + messages
    message = self.model.invoke(messages)
    return {'messages': [message]}

def take_action(self, state: AgentState):
    tool_calls = state['messages'][-1].tool_calls
    results = []
    for t in tool_calls:
        print(f"Calling: {t}")
        if not t['name'] in self.tools:      # check for bad tool name
            print("\n ....bad tool name....")
            result = "bad tool name, retry" # instruct LLM to retry if
        else:
            result = self.tools[t['name']].invoke(t['args'])
            results.append(ToolMessage(tool_call_id=t['id'], name=t['name'],
        print("Back to the model!")
    return {'messages': results}

```

```

In [6]: prompt = """You are a smart research assistant. Use the search engine to look up information.
You are allowed to make multiple calls (either together or in sequence). \
Only look up information when you are sure of what you want. \
If you need to look up some information before asking a follow up question,
"""

model = ChatOpenAI(model="gpt-3.5-turbo") #reduce inference cost
abot = Agent(model, [tool], system=prompt)

```

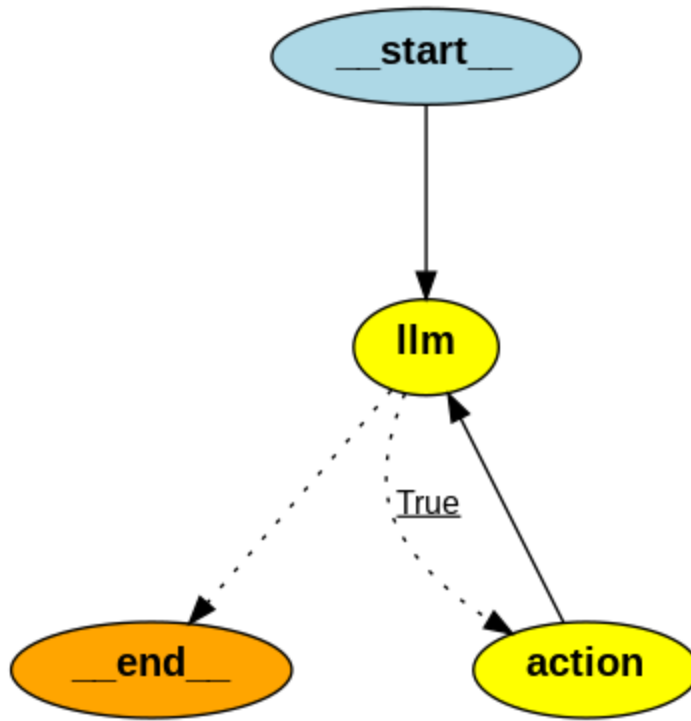
```

In [7]: from IPython.display import Image

Image(abot.graph.get_graph().draw_png())

```

Out [7]:



```
In [8]: messages = [HumanMessage(content="What is the weather in sf?")]  
result = abot.graph.invoke({"messages": messages})
```

Calling: {'name': 'tavily_search_results_json', 'args': {'query': 'weather in San Francisco'}, 'id': 'call_L0waGIuM75IXb7742hPrERIB'}
Back to the model!

```
In [9]: result
```

```

Out[9]: {'messages': [HumanMessage(content='What is the weather in sf?'),
  AIMessage(content='', additional_kwargs={'tool_calls': [{'id': 'call_L0waGIuM75IXb7742hPrERIB', 'function': {'arguments': '{"query":"weather in San Francisco"}', 'name': 'tavily_search_results_json', 'type': 'function'}]}, response_metadata={'token_usage': {'completion_tokens': 21, 'prompt_tokens': 153, 'total_tokens': 174}, 'model_name': 'gpt-3.5-turbo', 'system_fingerprint': None, 'finish_reason': 'tool_calls', 'logprobs': None}, id='run-27381d11-5455-40f3-96e9-84e02332ea1d-0', tool_calls=[{'name': 'tavily_search_results_json', 'args': {'query': 'weather in San Francisco'}, 'id': 'call_L0waGIuM75IXb7742hPrERIB'}])),
  ToolMessage(content='[{"url": "https://www.weatherapi.com/", "content": {"location": {"name": "San Francisco", "region": "California", "country": "United States of America", "lat": 37.78, "lon": -122.42, "tz_id": "America/Los_Angeles", "localtime_epoch": 1718022245, "localtime": "2024-06-10 5:24"}, "current": {"last_updated_epoch": 1718021700, "last_updated": "2024-06-10 05:15", "temp_c": 12.2, "temp_f": 54.0, "is_day": 0, "condition": {"text": "Overcast", "icon": "//cdn.weatherapi.com/weather/64x64/night/122.png", "code": 1009}, "wind_mph": 8.1, "wind_kph": 13.0, "wind_degree": 250, "wind_dir": "WSW", "pressure_mb": 1015.0, "pressure_in": 29.96, "precip_mm": 0.0, "precip_in": 0.0, "humidity": 93, "cloud": 100, "feelslike_c": 11.3, "feelslike_f": 52.3, "windchill_c": 10.8, "windchill_f": 51.4, "heatindex_c": 11.8, "heatindex_f": 53.3, "dewpoint_c": 10.0, "dewpoint_f": 50.1, "vis_km": 16.0, "vis_miles": 9.0, "uv": 1.0, "gust_mph": 9.0, "gust_kph": 14.5}}], {"url": "https://forecast.weather.gov/zipcity.php?inputstring=San+Francisco,CA", "content": "San Francisco CA 37.77°N 122.41°W (Elev. 131 ft) Last Update: 6:00 pm PDT Jun 7, 2024. Forecast Valid: 7pm PDT Jun 7, 2024-6pm PDT Jun 14, 2024 . Forecast Discussion . Additional Resources. Radar & Satellite Image. Hourly Weather Forecast. ... Severe Weather ; Current Outlook Maps ; Drought ; Fire Weather ; Fronts/Precipitation Maps ; Current ..."}, {"url": "https://forecast.weather.gov/MapClick.php?x=165&y=156&site=mtr&zm=x&map_x=165&map_y=156", "content": "2 Miles S San Francisco CA 37.77°N 122.44°W (Elev. 253 ft) Last Update: 1:27 pm PDT Jun 6, 2024. Forecast Valid: 8pm PDT Jun 6, 2024-6pm PDT Jun 13, 2024 . Forecast Discussion . Additional Resources. Radar & Satellite Image. Hourly Weather Forecast ... Severe Weather ; Current Outlook Maps ; Drought ; Fire Weather ; Fronts/Precipitation Maps ..."}, {"url": "https://weather.com/weather/tenday/l/San+Francisco+CA+USCA0987:1:US", "content": "Comfy & Cozy\nThat's Not What Was Expected\nOutside\nNo-Name Storms' In Florida\nGifts From On High\nWhat To Do For Wheezing\nSurviving The Season\nStay Safe\nAir Quality Index\nAir quality is considered satisfactory, and air pollution poses little or no risk.\nHealth & Activities\nSeasonal Allergies and Pollen Count Forecast\nNo pollen detected in your area\nCold & Flu Forecast\nFlu risk is low in your area\nWe recognize our responsibility to use data and technology for good. recents\nSpecialty Forecasts\n10 Day Weather-San Francisco, CA\nToday\nMon 18 | Day\nConsiderable cloudiness. Tue 19\nTue 19 | Day\nLight rain early...then remaining cloudy with showers in the afternoon. Wed 27\nWed 27 | Day\nOvercast with rain showers at times."}]', name='tavily_search_results_json', tool_call_id='call_L0waGIuM75IXb7742hPrERIB'),
  AIMessage(content='The current weather in San Francisco is overcast with a temperature of 54.0°F (12.2°C). The wind speed is 13.0 kph from the WSW direction, and the humidity is at 93%.', response_metadata={'token_usage': {'completion_tokens': 49, 'prompt_tokens': 1099, 'total_tokens': 1148}, 'model_name': 'gpt-3.5-turbo', 'system_fingerprint': None, 'finish_reason': 'stop', 'logprobs': None}, id='run-e08e5b60-fabc-4b92-aeb7-e87682c0addb-0')]]

```

```
In [10]: result['messages'][-1].content
```

```
Out[10]: 'The current weather in San Francisco is overcast with a temperature of 54.0°F (12.2°C). The wind speed is 13.0 kph from the WSW direction, and the humidity is at 93%.'
```

```
In [11]: messages = [HumanMessage(content="What is the weather in SF and LA?")]
result = abot.graph.invoke({"messages": messages})
```

```
Calling: {'name': 'tavily_search_results_json', 'args': {'query': 'weather in San Francisco'}, 'id': 'call_lkr5RWNwB5xy6tqH4Q270cBV'}
```

```
Calling: {'name': 'tavily_search_results_json', 'args': {'query': 'weather in Los Angeles'}, 'id': 'call_43in9jCICtpspXGFik73h6ww'}
```

```
Back to the model!
```

```
In [12]: result['messages'][-1].content
```

```
Out[12]: 'The current weather in San Francisco is overcast with a temperature of 54.0°F. The wind speed is 13.0 kph coming from WSW, and the humidity is at 93%.\n\nIn Los Angeles, the current weather is also overcast with a temperature of 62.1°F. The wind speed is 3.6 kph coming from the north, and the humidity is at 80%.'
```

```
In [13]: # Note, the query was modified to produce more consistent results.
# Results may vary per run and over time as search information and models change
```

```
query = "Who won the super bowl in 2024? In what state is the winning team located? What is the GDP of that state? Answer each question."
messages = [HumanMessage(content=query)]
```

```
model = ChatOpenAI(model="gpt-4o") # requires more advanced model
abot = Agent(model, [tool], system=prompt)
result = abot.graph.invoke({"messages": messages})
```

```
Calling: {'name': 'tavily_search_results_json', 'args': {'query': 'who won the Super Bowl 2024'}, 'id': 'call_mbIBJ6gNk9SurZVtkQ0qzm6c'}
```

```
Back to the model!
```

```
Calling: {'name': 'tavily_search_results_json', 'args': {'query': 'Kansas City Chiefs headquarters location'}, 'id': 'call_x6xIpSJ831XnyQXjWJbI37co'}
```

```
Back to the model!
```

```
Calling: {'name': 'tavily_search_results_json', 'args': {'query': 'GDP of Missouri 2024'}, 'id': 'call_ENCBsYAWZV5D3R60hN1XDAlt'}
```

```
Back to the model!
```

```
In [14]: print(result['messages'][-1].content)
```

1. ****Who won the Super Bowl in 2024?****
 - The Kansas City Chiefs won the Super Bowl in 2024, defeating the San Francisco 49ers with a score of 25-22.
2. ****In what state is the winning team's headquarters located?****
 - The Kansas City Chiefs' headquarters is located in Missouri.
3. ****What is the GDP of that state?****
 - The real gross domestic product (GDP) of Missouri in 2023 was approximately \$344.12 billion USD. The GDP for 2024 has not been specifically listed, but it is likely to be similar or slightly higher.

In []: